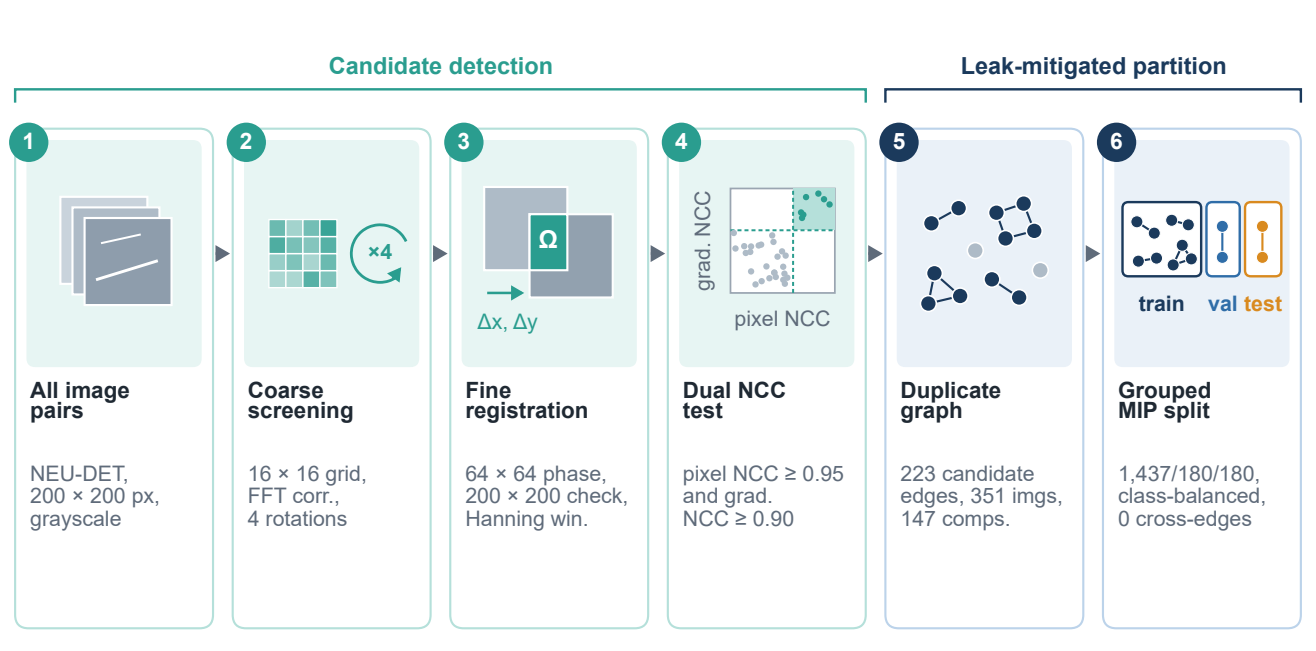

Candidate detection
Leak-mitigated partition
1
2
3
4
5
6
×4
Ω
Δx, Δy
grad. NCC
pixel NCC
train
val
test
All image pairs
NEU-DET, 200 × 200 px, grayscale
Coarse screening
16 × 16 grid, FFT corr., 4 rotations
Fine registration
64 × 64 phase, 200 × 200 check, Hanning win.
Dual NCC test
pixel NCC ≥ 0.95 and grad. NCC ≥ 0.90
Duplicate graph
223 candidate edges, 351 imgs, 147 comps.
Grouped MIP split
1,437/180/180, class-balanced, 0 cross-edges